

Let $X_1, X_2, \dots, X_n$ be independent random variables such that $X_i \in [a_i, b_i]$ almost surely for all i . We define

$$S_n = \sum_{i=1}^n X_i.$$

Then, $\forall \varepsilon > 0$,

$$Pr(S_n - \mathbb{E}[S_n] \geq \varepsilon) \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and similarly,

$$Pr(S_n - \mathbb{E}[S_n] \leq -\varepsilon) \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Equivalently, for $\tilde{X} = S_n/n$,

$$Pr(\tilde{X} - \mathbb{E}[\tilde{X}] \geq \varepsilon) \leq \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$